



DARPA's radio challenge!

DARPA has challenged participants to develop radio that prevents wireless congestion
digit.in/Darpachallenge



Remember BB-8?

The insides of BB-8 from Star-Wars and how it works
digit.in/StarwarsBB8

Bazaar

AMD RX 480

AMD has hit it out of the park with the RX 480

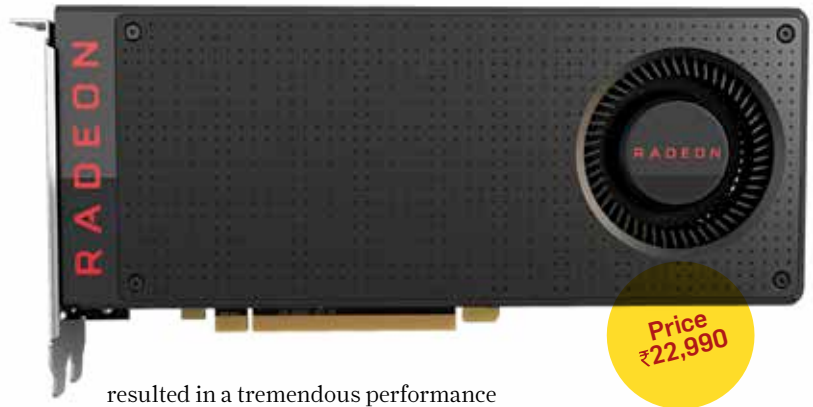
The AMD RX 480 is driving AMD's return to the desktop graphics arena. We say desktop because they've been ruling the console arena for quite a while as both Microsoft's Xbox One and Sony's PS4 run on AMD's graphics solutions and even their upcoming console refreshes are reportedly carrying Polaris based GPUs. Priced at Rs.20,990 and Rs.22,990 for the 4 GB and 8 GB VRAM SKUs, the AMD RX 480 might not seem to be competitively priced in India. After all, these aren't the launch prices as AMD dropped Indian prices for the RX 480 by Rs.2,000 within two days of the card's launch and even further about a month later.

Performance

We started off with 3DMark managing to score 10511 in Fire Strike, 5201 in Fire Strike Extreme and 2691 in Fire Strike Ultra. Unigine Heaven 4.0 scored 1635 with the RX 480 which is quite close to the GTX 970 and definitely better than the 960.

Then came Shadow of Mordor. While the game is a bit old it still manages to tax graphics cards while running at 4K resolutions and that is exactly the case here. The game's internal benchmark runs effortlessly at 1080p but cranking it up to 1440p brings the FPS below the 60 FPS threshold and 4K takes it down even further. However, the game is playable at 4K with very few high-NPC scenarios actually causing the FPS scores to drop below playable levels.

On the other hand, Rise of the Tomb Raider was a completely different scenario. Using our extreme preset even at 1080p the game ran a little below 60 FPS. Needless to say, 1440p and 4K was even lower with the game managing roughly 13 FPS on average. We are quite eager to see how well the game performs with two RX 480 in CrossFireX. Having a closer look at the frame times, it seems that out of the tree sequences, the first two run quite smoothly. Benchmarking DOOM was quite interesting, especially since we had the option of switching between OpenGL and Vulkan APIs. The game ran quite smoothly throughout the entire benchmark sequence on either APIs at 4K. The only place we saw a bit of delay was while panning across wide open spaces with lots of objects far in the background. However, the complete eye-opener was the performance difference between the two APIs. Running the benchmarks with Vulkan



resulted in a tremendous performance increase. Throughout the benchmark across all three resolutions, we saw an increment of approximately 35%. Since Vulkan is heavily based on AMD's Mantle, this makes us wonder how Mantle would have performed had it been adopted by the industry widely.

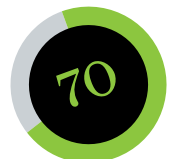
Verdict

AMD has truly hit it out of the park with Polaris. The new architecture coupled with the transition to 14nm FinFET has resulted in profound power savings and performance increase. The card at launch did draw an excess amount of power from the PCIe slot instead of the power connectors but that was resolved in driver version 16.7.1 and there has been a good amount of performance increment with Vulkan as well.

The only factor that throws a spanner in the gears is that of pricing. While it may be \$199 in overseas markets, there are countries like India and Australia where the FOB isn't the same as overseas markets. Resulting in a price gain, add to that a convoluted distribution hierarchy and we have an even greater price rise. However, this affects both AMD and NVIDIA so the RX 480 and its counterpart, the GTX 1060, are in the same price bracket in India. Both of these cards are meant to be the sweet spot cards but that's going to change, at least for India.

The RX 480 is a great card, it runs almost all current gen AAA titles at 1080p at max with no issues. Even 1440p isn't that taxing, and if you know which settings to tweak then you'll enjoy 1440p gaming with no visual difference than 1080p. However, when it comes to 4K, the RX 480 needs a helping hand from another RX 480 and here's where the AMD card has an advantage.

Mithun Mobandas



Performance..... 70
Build 75
Value for Money 65

Specifications

RX 480 Polaris; **Stream Processors:** 2304; **Compute Units:** 36; **Core clock:** 1120 MHz; **Boost clock:** 1266 MHz; **VRAM:** 4 / 8 GB GDDR5; **Memory clock:** 7 Gbps (4GB) / 8 Gbps (8GB); **Memory bus width:** 256-bit; **TDP:** 150 W; **Power Connectors:** 1x 6-PIN

Contact

AMD
Contact: <https://support.amd.com/en-us/contact>
Website: www.amd.com